LGST-Drop : Label-Guided Structural Dropout for Spatial-Temporal Convolutional Neural Networks

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***Abstract*—Region dropout regularization strategies have been proven highly effective in improving the generalization performance of Convolutional Neural Networks (CNNs) in a variety of computer vision tasks including image classification, object detection, semantic segmentation, because these strategies enabling models to focus on a wider range of image region information. However, for action recognition, models need to be able to extract not only useful spatial information, but also important temporal and motion information, which cannot be satisfied by traditional regularization strategies. In this paper, we propose a spatiotemporal dropout strategy to meet the need for regularization in spatial-temporal CNNs. We call it Label Guided Spatial-Temporal Drop (LGST-Drop) which not only provides effectively structured dropout in the spatial dimension but also regularizes motion information in the temporal dimension. In addition, LGST-Drop's Mask is guided by the predicted categories of the model itself which we called Temporary Labels. Extensive experiments on several standard datasets from action recognition domains show the usefulness of the proposed technique in comparison with the previous methods and its state-of-the-art variants algorithms.**

I. Introduction

Deep convolutional neural networks have a wide range of applications in computer vision. Many fruitful networks[1], [2], [3], [4], [5], [6] have been designed and succeeded. In order to solve the overfitting problem caused by the increasing number of convolutional neural network parameters, many regularization approaches have been proposed, such as weight decay[7], [8], [9], data augmentation[10], [11], dropout series methods[12], [13], [14], [15], [16], which are often effective in basic computer vision tasks. However, video-based action recognition[17], [18], [19], [20] differs from other image-based tasks because it introduces temporal dimensional information in the model. And the temporal and spatial dimensions are often not in an equivalent relationship, which leads to these regularization methods not so effective in action recognition networks.

In the spatial dimension, applying the ordinary dropout[12] strategy to the convolutional layer does not work well as it does on the fully connected layer (FC-layer), because the information dropped in the convolutional layer can easily be repaired by the regions near the dropped locations. To solve this problem, some structured drop methods have been proposed, such as Cutout[21], CutMix[22], DropBlock[14], because they are random dropout strategies, over or under regularization problems still exist. Therefore some non-random drop strategies are proposed. AttentionDrop[15] combines dropout with an attention mechanism to improve the efficiency of dropout, CorrDrop[23] selects the feature units by the values of features correlation in the spatial dimension. TargetDrop[16] decides the feature unit regions to be dropped by the attention mechanism. These methods discard the network neural regions with high activation values in the spatial dimension of the image. The disadvantage is that they ignore the inherent information in the channel dimension which is proven to be critical for the final result.

Fig. 1: CutMix[22], naive Dropout[12], DropBlock[14] and LGST-Drop in video frames or feature maps. The blue regions denote the regions to be masked. The red frames denote the important frames.

In the temporal dimension, the number of frames T takes relatively smaller values compared to the spatial scale (T is generally taken as 8/16, and HW is generally taken as 224/256). Therefore, it is obviously unreasonable to carry out an unified drop strategy at the spatial and temporal scales. On the other hand, giving up dropping in the temporal dimension means ignoring temporal information regularization, that is not feasible too.

To solve the above two problems, in this paper, we propose a regularization strategy Label Guided Spatial-Temporal Dropblock (LGST-Drop) for video input data and spatial-temporal networks[24], [25], [26], [27], [28]. As shown in Figure 1, our approach can explicitly discard feature map units on both spatial and temporal regions relating to the target labels, and implicitly measure the importance of different channels of every frame's feature map. Specifically, before each forward propagation, we use the model to predict the training videos' labels which are named Temporary Labels. And then get the activation maps of the middle features from Temporary Labels

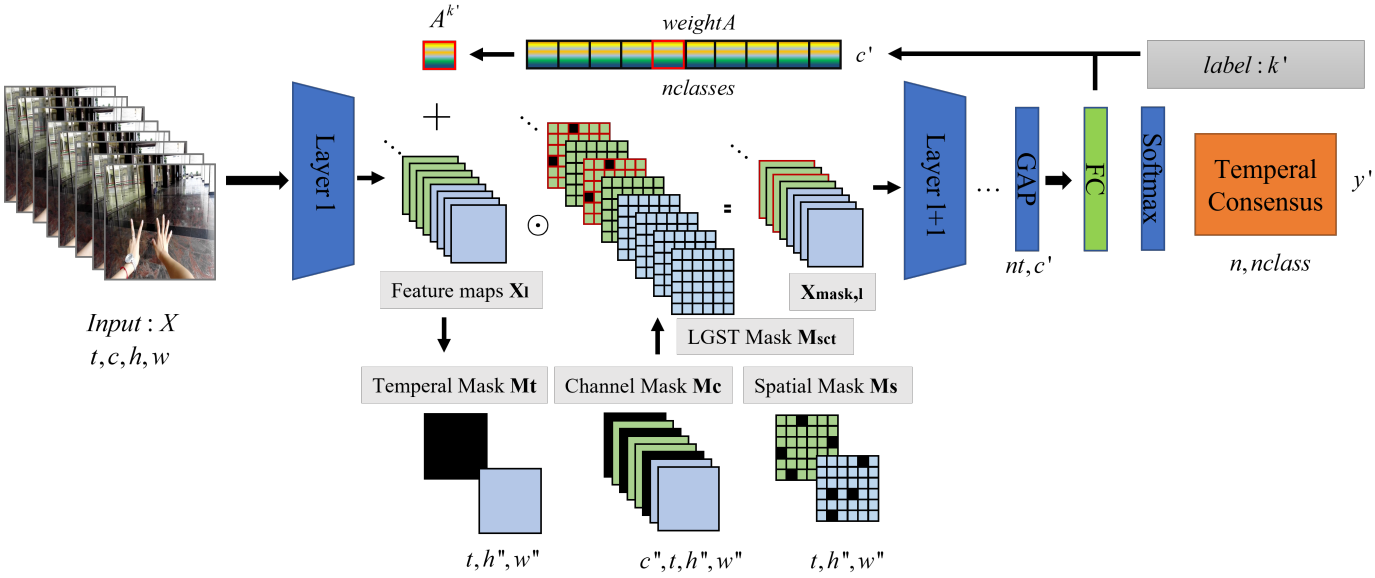

Fig. 2: The pipeline of LGST-Drop. Firstly, before each forward propagation, we use the model to predict the input temporary label k'. Next, the weight $A^{k'}$ which related to k' is extracted from the penultimate FC-layer of the network. Then, we compute the LGST-Drop Mask M^{sct} from $A^{k'}$ and mid feature maps X_l. Finally, the mask M^{sct} is applied to the original feature maps X_l by the multiplication operation.

and the weights of the fully connected layer. The spatial drop mask $\mathbf{M}^{(\mathbf{s})}$is obtained by zeroing the largest values in the activation maps. Because there exist quite a wealth of motion semantic information in the channel and time dimensions as well, we compute the channel dimension drop mask $\mathbf{M}^{(\mathbf{c})}$ and the temporal dimension drop mask $\mathbf{M}^{(\mathbf{t})}$ from the label-related weights in Temporary Labels. Finally, we combine the three masks to obtain the final spatiotemporal drop mask $\mathbf{M}^{(\mathbf{sct})}$, which can effectively drop the distinctive and shortcut feature units of the input videos from three dimensions and forces the network to learn more critical information from other regions, channels and frames. Experiments show that the LGST-Drop in this paper can significantly improve the performance of CNNs in action recognition compared with other regularization methods.

Our contributions are summarized as follows:

- We propose a novel non-random regularization method which is guided by Temporary Labels of samples, it can solve the problem of unexpected results caused by random drops.
- We propose a structural dropout strategy for video data and spatial-temporal CNNs, it drops units from three dimensions: sptial, channel and temporal, it can solve the problem caused by dimensional asymmetry.
- Our method achieves a better regularization effect compared with the other state-of-the-art methods and applies to different architectures for action recognition tasks.

II. Related Works

A. Spatial Temperal Models

The key difference between action recognition[29], [30], [31] and image recognition[1], [3] is the requirement of temporal information modeling. For example, to distinguish between a finger swipe to the left and a swipe to the right, reversing the order of video frames will yield opposite results. So temporal modeling is essential in action recognition. Existing efficient video understanding methods are based on 2DCNNs[26], [27], [32] , 3DCNNs[33], [34], [35], [36] and Transformer[37]. Because 2DCNNs cannot directly process the time dimension information well, it needs to interact with some other methods such as optical flow[38], [39], temporal shift convolution (TSM)[26] which can model temporal information. 3DCNNs can comprehensively learn useful information in space and time dimensions, but the greater computational cost makes it difficult to deploy on low-cost devices. The Transformer-based approach applies the Transformer[40] architecture to video data by learning the spatiotemporal features of the videos from a series of frame-level image patches. This paper is about the method of convolution regularization in the field of action recognition, so the method based on Transformer is not discussed, the baseline is the architecture based on 2DCNNs and TSM, as it is computationally moderate and does not require other additional data such as optical flow.

B. Dropblock Methods

Since Dropout was proposed to improve the performance of CNNs by alleviating the overfitting problem, a series of related variants have been proposed such as DropConnect[41], SptialDropout[13], DropPath[42],

Method	M.O.	Type
Dropout	Points	random
SptialDrop	Channels	random
AttentionDrop	Areas	guided
DropPath	Layers	random
DropBlock	Areas	random
LGST-Drop	Areas,Channels,Tp.	guided

TABLE I: Different drop method masks characteristic analysis. M.O. denotes Mask Object, Points means drop points, Channels means drop channels, Areas means drop space area in feature maps, Tp. means drop in temporal dimension.

DropBlock[14], AttentionDrop[15], CorrDrop[23] and DropFilter[43]. They can be generally classified into two categories, random dropout strategies and guided dropout strategies. AttentionDrop and CorrDrop both belong to the latter, they select the neuronal regions to drop according to attention mechanism and feature correlations respectively. Our LGST-Drop is similar to them, but there are two differences: Firstly, the guiding strategy in our method is based on the Temporary Labels. Secondly, we set different drop methods for different dimensions, and the general procedure is shown in Figure 2.

III. Method

In this section, we first give the general formulation of Dropout family, and then describe the computation procedure of our LGST-Drop Masks in spatial, channel and temporal dimensions.

A. Notations

For ease of understanding, we give some definitions and notations necessary. A typical CNNs with N layers for action classification denoted as a function $f(\mathbf{X}, y) : \mathbb{R}^{C\times T\times H\times W} \rightarrow \mathbb{R}^K$, where $\mathbf{X}, y$ is the input of CNNs and label with K classes respectively. Suppose the input of l layer of CNNs is $\mathbf{X}^{l-1}$ and the output this layer is $\mathbf{X}^l$, every layer in CNNs can be expressed as $\mathbf{X}^l = \sigma\left(\mathbf{W}^l\mathbf{X}^{l-1} + b^l\right)$, where $\mathbf{W}^l$ is the parameter matrix, b^l is bias vector, $\sigma(\cdot)$ is the compound operators(e.g. Pooling, Norm, Relu). Suppose $\mathbf{Z} \in \mathbb{R}^{C'\times T\times H'\times W'}$ is the output of last convlutional layer. Then, global average pooling is performed to reduce the sptial and temporal dimensions and we get the resulting vector $z \in \mathbb{R}^{C'\times 1}$, finally, fed z into the FC-layer with weight matrix $\mathbf{A} \in \mathbb{R}^{C'\times K}$ and softmax the result to predict the probability $y'_{k'}$ for k'. The procedure can be expressed as equation 1 and 2.

$$S^k = z\alpha^k = \sum_{c=1}^{C'} \mathrm{A}_{c,k} \frac{1}{TH'W'} \sum_{t=1}^{T}\sum_{h=1}^{H'}\sum_{w=1}^{W'} Z_{c,t,h,w} \tag{1}$$

$$y'_k = P\left(y = k' \mid z\right) = \exp\left(S^{k'}\right) / \sum_{k=1}^{K} \exp\left(S^k\right) \tag{2}$$

B. Dropout Formula

Dropout method is applied in the feature map of the network, its general formula is equation 3 and 4.

$$\tilde{\mathbf{X}}^l = \mathbf{M}^l \odot \mathbf{X}^l / \theta \tag{3}$$

$$\mathbf{M}^l_{c,t,h,w} \sim \mathrm{Ber}(\theta), \quad \theta \in [0, 1] \tag{4}$$

where θ is the keep rate, $\odot$ denote Hadamard product, $\mathbf{M}^l$ is the drop mask in which every element value is sampled from a Bernoulli distribution. As show in Table I, the differences between the various Drop methods are due to the different calculation methods and targets of their masks. The LGST-Drop mask $\mathbf{M}^l$ in this paper is not only applied to the spatial region, but also to the channel and temporal dimensions.

C. LGST-Drop

The pipeline of LGST-Drop is shown in Fig. 2. Firstly, we extracted the weights corresponding to the Temporary Labels from the FC-layer of the network. Then, compute spatial masks, channel masks and temporal masks by convolution layers' feature maps and the label-weights. Finally, the masks are applied to the original feature maps and replace the original. In this section, we will describe the three masks and the whole procedure of LGST-Drop algorithm in detail.

Label Guided Spatial Mask. Intuitively, we would like the model to be robust by focusing on a wider range of regions and obtaining as much useful information from these regions as possible, rather than just focusing on those areas that can easily distinguish the objects from each other. Because this would bias the model towards shortcuts and therefore become one-sided. Although DropBlock can solve this problem to a certain extent, but since it is a random drop strategy, the drop area may not be the one that contains shortcut information. To solve this problem, we introduce a temporary Label Guided mechanism to make the model more effective in dropping the regions that are closely related to the Temporary Labels, and then force the model to learn more general valuable information from other more extensive regions for category classification.

From section III-A, we know the FC-layer's weight matrix is $\mathbf{A} = \{\alpha^1, \alpha^2, \ldots, \alpha^K\} \in \mathbb{R}^{C'\times K}$, the input of global pooling layer is $\mathbf{Z} \in \mathbb{R}^{C'\times T\times H'\times W'}$. Suppose the Temperal Label is k', we can get the label-weight $\alpha^{k'} \in \mathbb{R}^{C'\times 1}$ from $\mathbf{A}$, which can be considered as channels' important scores of the feature map $\mathbf{Z}$ for class k'. We can obtain the saptial pixel-weighted map for each frame from equation 5 and 6.

$$\mathbf{Z}' = \mathrm{Reshape}(\mathbf{Z}) \in \mathbb{R}^{NT\times C'\times H'\times W'} \tag{5}$$

$$\mathbf{G}^{k'}_S = \sum_{c=1}^{C'} a^{k'}_c \mathbf{Z}'_{c,:,:} \tag{6}$$

Since we only need to focus on the spatial dimension at the moment, the input temporal dimension T and batchsize N are combined into one dimension, so that the spatial pixel-weight maps are different for different frames. This is more logical than Cutout or CutMix as shown in Fig. 1. Here,

the spatial pixel-weighted map $\mathbf{G}_S^{k'}$ can be considered as a set of $H' \times W'$ elements if ignore the first dimension NT, which represents the pixels' importance of the feature map in the spatial dimension. Suppose $\mathbf{D}_{n_s}^s$ is the set of the most important n_s elements. We can obtain point spatial mask $\mathbf{M}^{(\mathrm{ps})}$ from equation 7.

$$\mathbf{M}_{h,w}^{(\mathrm{ps})} = \begin{cases} 0, \mathbf{G}_{h,w}^{k'} > \min\left(\mathbf{D}_{n_s}^s\right) \\ 1, \mathbf{G}_{h,w}^{k'} < \min\left(\mathbf{D}_{n_s}^s\right) \end{cases} \tag{7}$$

We assume the channel of a mid convolutional layer's feature map is C'', if it is not equal to C', it's essential to adjust $\alpha^{k'} \in \mathbb{R}^{C' \times 1}$ in channel dimension. In this situation, the equation 6 is turned into equation 8.

$$\mathbf{G}_S^{k'} = \sum_{c=1}^{C''} \left(\sum_{j=(c-1)\times\left[\frac{C'}{C''}\right]+1}^{c\times\left[\frac{C'}{C''}\right]} a_j^{k'} \right) \mathbf{X}_{c,:,:}^l \tag{8}$$

In equation 7, the value of n_s get from equation 9 where γ denotes the dropblock size, θ denotes the value of keep rate. H'', W'' denote the height and width of $\mathbf{X}^l$.

$$n_s = \frac{(1-\theta)}{\gamma^2} \times H''W'' \tag{9}$$

$$\mathbf{M}^{(\mathrm{s})} = \mathbf{Expand}(\mathbf{kernel_size} = \gamma, \mathbf{input} = \mathbf{M}^{(\mathrm{ps})}) \tag{10}$$

After obtaining the point spatial mask $\mathbf{M}_{h,w}^{(\mathrm{ps})}$, we need to expand the points to blocks for preventing the network from estimating and complementing the dropped points' values. Finally, we can get the Label Guided Spatial Mask $\mathbf{M}^{(\mathrm{s})}$ from equation 10.

Label Guided Channel Mask. Although $\mathbf{G}_S^{k'} \in \mathbb{R}^{H'' \times W''}$ can represent the importance in sptial dimension for every frame. But because it is the result of weighted summation of all channels, so that it's identical for all channels which is irrational. For example, if the pixel-weighted map $\mathbf{G}_S^{k'}$ has a largest value near location (h_1, w_1), the largest value maybe near location (h_1, w_1) in channel c_i of $\mathbf{X}^l$ too, because channel c_i have larger weight in equation 8. But the maximum value location maybe far away from (h_1, w_1) for another channel c_j since smaller weight of this channel, and its drop region will near the location (h_1, w_1) same as channel c_i.

Therefore, it is essential to build a mask that can distinguish the importance of different channels. We can get the label-weight $\alpha^{k'} \in \mathbb{R}^{C' \times 1}$ from **A** III-A. Suppose $\mathbf{D}_{n_c}^c$ is the set of the top n_c significant values in $\alpha^{k'}$, we can obtain the Label Guided Channel Mask $\mathbf{M}^{(\mathbf{c})}$ from equation 11 in where n_c can be got from equation 12.

$$\mathbf{M}_c^{(\mathbf{c})} = \begin{cases} 0, \alpha_c^{k'} > \min\left(\mathbf{D}_{n_c}^c\right) \\ 1, \alpha_c^{k'} < \min\left(\mathbf{D}_{n_c}^c\right) \end{cases} \tag{11}$$

$$n_c = \frac{1}{\mu} \times C'' \tag{12}$$

where μ is a new hyperparameter.

Algorithm 1: LGST-Dropblock

Input : feture map of a layer $\mathbf{X_l} \in \mathbf{R}^{\mathbf{C''\times T\times H''\times W''}}$, drop block_size $\mathbf{r}$, input data lable $\mathbf{k'}$, weight of penultimate FC-layer $\mathbf{A} \in \mathbf{R}^{\mathbf{C'\times N_classes}}$, output feture of last layer $\mathbf{Z} \in \mathbf{R}^{\mathbf{C'\times T\times H'\times W'}}$, drop elements number in space $\mathbf{ns}$, drop channel number $\mathbf{nc}$, drop frame number $\mathbf{nt}$.

Output: $\mathbf{X_{mask,l}}$

STEP : compute label weight :
$\mathbf{A^{k'}} \leftarrow \mathbf{A}, \mathbf{k'}$;
if $C' \neq C$ **then**
  $\mathbf{A_w}$ = `Averagepool` ($\mathbf{A^{k'}}$, kernel=C'/C'', stride=C'/C'')
else $\mathbf{A_w} = \mathbf{A^{k'}}$

STEP : compute spatial mask for all frames:
for $t \leftarrow 0$ **to** T; `// T is frame numbers`
do
  $\mathbf{x_t} \leftarrow$ `ExtractFrame` ($\mathbf{X_l}$, t)
  $\mathbf{x_t} \leftarrow$ `Reshape` ($\mathbf{x_t}$, shape=$(H'' \times W'', C'')$);
  $\mathbf{m_s} \leftarrow$ `Mmul` ($\mathbf{x_t}$, $\mathbf{A_w}$) $\mathbf{D_s} \leftarrow$ `Topk` ($\mathbf{m_s}$, n=**ns**)
  $\mathbf{m_s} \leftarrow \mathbf{m_s} <$ `Min` ($\mathbf{D_s}$) ;
  $\mathbf{m_s} \leftarrow$ - `Maxpool` (-$\mathbf{m_s}$, kernel_size=**r**);
$\mathbf{M_s}$ = `ConCat` ($\mathbf{m_1}$, $\mathbf{m_2}$,...,$\mathbf{m_T}$);

STEP : compute channel mask :
$\mathbf{D_c} \leftarrow$ `Topk` ($\mathbf{A_w}$, n=**nc**);
$\mathbf{M_c}$ = `Ones` (shape=(C'', T, H'', W''));
for c *in* $\mathbf{D_c}$ **do**
  $\mathbf{M_c}[c, :::] \leftarrow 0$;

STEP : compute temporal mask :
$\mathbf{F} \leftarrow$ `Averagepool` ($\mathbf{X_l}$, dim=(h,w));
$\mathbf{w_f}$ = `Mmul` ($\mathbf{F}$, $\mathbf{A_w}$) ; `//` $\mathbf{A_w}$ `shape` $C'' \times 1$
$\mathbf{D_t} \leftarrow$ `Topk` ($\mathbf{w_f}$, n=**nt**);
$\mathbf{M_t}$ = `Ones` (shape=(C'', T, H'', W''));
for t *in* $\mathbf{D_t}$ **do**
  $\mathbf{M_t}[:, t, ::] \leftarrow 0$;

STEP : merge Ms,Mc,Mt
$\mathbf{M_{s,c,t}}$ = `LogicOr` ($\mathbf{M_s}$,$\mathbf{M_c}$,$\mathbf{M_t}$)
if *Training* **then**
  $\mathbf{X_{mask,l}} \leftarrow$ `Mul` ($\mathbf{X_l}, \mathbf{M_{s,c,t}}$)
return $\mathbf{X_{mask,l}}$

Label Guided Temporal Mask. From the previous sections we know the $\mathbf{M}^{(\mathrm{s})}$ masks in the spatial dimension and $\mathbf{M}^{(\mathbf{c})}$ masks in the channel dimension, but this is not enough for video data and spatial temporal models used for action recognition. Because it is proved that the information in temporal dimension plays a very important role in action recognition. So it is essential to introduce some regularization techniques to temporal dimension. We propose a temporal dropout mechanism similar to what in channel dimension.

$$\mathbf{Z}'' = \frac{1}{H''W''} \sum_{h=1}^{H''} \sum_{w=1}^{W''} \mathbf{X}^l(c,t,h,w) \tag{13}$$

$$\mathbf{G}_T^{k'} = \sum_{c=1}^{C''} \left(\sum_{j=(c-1)\times\left[\frac{C'}{C''}\right]+1}^{c\times\left[\frac{C'}{C''}\right]} a_j^{k'} \right) \mathbf{Z}''_{c,:} \tag{14}$$

$$\mathbf{M}_t^{(t)} = \begin{cases} 0, \mathbf{G}_t^{k'} > \min\left(\mathbf{D}_{n_t}^t\right) \\ 1, \mathbf{G}_t^{k'} < \min\left(\mathbf{D}_{n_t}^t\right) \end{cases} \tag{15}$$

$$n_t = \text{ segments } \times \frac{1}{\beta} \tag{16}$$

Firstly, we compute temporal matrix $\mathbf{Z}'' \in \mathbb{R}^{C''\times T\times 1\times 1}$ by max average pooling in spatial dimension of $\mathbf{X}^l$ as shown in equation 13. Then, get label guided temporal weight map $\mathbf{G}_T^{k'} \in \mathbb{R}^{1\times T\times 1\times 1}$. Suppose $\mathbf{D}_{n_t}^t$ is the set of top n_t important elements. We can obtain the Label Guided Temporal Mask $\mathbf{M}^{(t)}$ from equation 15 in where n_t is computed from equation 16, and β is a new hyperparameter.

Label Guided Spatial Temporal Dropblock After obtaining the above three masks $\mathbf{M}^{(s)}$, $\mathbf{M}^{(c)}$,$\mathbf{M}^{(t)}$, we expand their shape to the size same as $\mathbf{X}^l$, taking logic_or operation for them, and the result is final LGST-Drop mask $\mathbf{M}^{(sct)}$. Algorithm 1 describes the detailed procedure of LGST-Drop.

IV. Experiments

In this section, we present the implementation details of our experiments and report the performance of our method. We compare our method with other dropout-based methods on the Something-Something V2[44] and Jester datasets[45], then we apply our method on different networks and report the results. Finally, we analyse the hyperparameters and demonstrate the visualization results.

A. Dataset and Implementation Details

Dataset. We evaluate our method on the Something-Something V2[44] and Jester datasets[45]. The Something-Something V2[17], [18], [19] is a collection of humans performing actions with everyday objects, including 174 categories, 168,913 training videos, 24,777 validation videos, and 27,157 test videos. Jester[20], [24], [25] is a third-person view gesture dataset that has potential uses in human-computer interaction, with 27 categories, containing 118,562 videos, 14,787 validation videos, and 14,743 test videos. It should be noted that in order to improve the efficiency of the experiments, we use subsets of the Something-Something V2 dataset in part experiments which only contain the first 12 categories (SS-V2-12).

Implementation Details. We experimented with the video action recognition tasks by following the same policies mentioned in TSM[26]. The models were trained on two NVIDIA Tesla A100 GPUs. We used SGD as optimizer with a momentum of 0.9 and a weight decay of 5×10^{-4}. The batchsize was set as N = 16 and the number of segments was set as T = 8. We started with a learning rate of 0.005 and reduced it by a factor of 10 at 20, 40 epochs and stopped at 70 epochs. We utilized the three-crop strategy following [26], [20], [28] for inference.

Method	SS-V2		Jester	
	top1	Δtop1	top1	Δtop1
ResNet50-TSM-dropout[12]	59.21	-	94.48	-
ResNet50-TSM-MixUp[46]	59.29	+0.08	94.42	-0.06
ResNet50-TSM-CutMix[22]	59.34	+0.13	94.47	-0.01
ResNet50-TSM-DropBlock[14]	59.55	+0.34	94.52	+0.04
ResNet50-TSM-AttentionDrop[15]	59.72	+0.51	94.67	+0.19
ResNet50-TSM-TargetDrop[16]	59.64	+0.43	**94.78**	**+0.30**
ResNet50-TSM-LGST	**59.83**	**+0.62**	94.76	+0.27
MB-V2-TSM-dropout[12]	55.51	-	95.04	-
MB-V2-TSM-MixUp[46]	55.62	+0.11	95.12	+0.08
MB-V2-TSM-CutMix[22]	55.69	+0.18	95.15	+0.11
MB-V2-TSM-DropBlock[14]	55.82	+0.31	95.21	+0.17
MB-V2-TSM-AttentionDrop[15]	55.92	+0.41	95.33	+0.29
MB-V2-TSM-TargetDrop[16]	55.91	+0.40	95.35	+0.31
MB-V2-TSM-LGST	**56.04**	**+0.53**	**95.39**	**+0.35**
BNIception-TSM-dropout[12]	54.21	-	94.71	-
BNIception-TSM-MixUp[46]	54.25	+0.04	94.66	-0.05
BNIception-TSM-CutMix[22]	54.37	+0.16	94.72	+0.01
BNIception-TSM-DropBlock[14]	54.42	+0.21	94.61	-0.10
BNIception-TSM-AttentionDrop[15]	54.49	+0.28	94.79	+0.08
BNIception-TSM-TargetDrop[16]	54.51	+0.30	94.82	+0.11
BNIception-TSM-LGST	**54.65**	**+0.44**	**94.87**	**+0.16**

TABLE II: Comparison against the results of different state-of-the-art dropout-based regularization methods on TSM with different backbone for classification accuracy of on Something-Something V2 and Jester. The paprameter α in MixUp and CutMix is set 0.2 following [47], the drop rate of dropout which only applied to the FC-layer is set 0.5 following [26], [29], the drop rate of AttentionDrop is set to 0.1 following [15], the drop rate and dropblock size of TargetDrop is set to 0.15 and 5 respectively following [16], the parameters in LGST-Drop are set to droprate = 0.15, dropblock size λ = 7, μ = 8, β = 2. Each experiment is conducted three times and we report the mean top-1 accuracy on validation datasets.

B. Comparisons with the State-of-the-Art

We compare the regularization effect with the other state-of-the-art methods including CutMix[22], MixUp[46], and other dropout-based methods[12], [14], [15], [16] on temporal shift network(TSM)[26] with different backbones such as ResNet-50[1], MobileNetv2[48], BNInception[2] in Table II. The table is split into three parts. In the top part, we present the results of all the regularization methods applied on TSM with the backbone of ResNet50. The middle part of the table is the results with the backbone of MobileNetv2, and the bottom part is the results with the backbone of BNInception. In all three parts, the results show that our method outperforms all the other methods on the Something-Something V2[44] dataset and is better than all other methods except TargetDrop[16] on the Jester[45] dataset. We can find that the improvement by applying regularization methods on the Something-Something V2 dataset is better than that on the Jester data, which we believe the reason is that the Something-Something V2 dataset

Method	top1	Δtop1	top5	Δtop5
ResNet-50	59.19	-	85.29	-
S-Dropblock	59.54	+0.35	85.50	+0.21
C-Dropblock	59.30	+0.11	85.31	+0.02
T-Dropblock	59.45	+0.26	85.55	+0.26
SC-Dropblock	59.77	+0.58	85.69	+0.40
SCT-Dropblock	59.83	+0.64	85.81	+0.52

TABLE III: Ablation experiment of the three masks of LGSP-Drop on Something-Something V2. ResNet-50 indicates the state when no drop is performed, S-Dropblock indicates spatial dropblock only, C-Dropblock indicates channel dropblock only, T-Dropblock indicates temporal dropblock only, SC-Dropblock indicates the condition when spatial and channel dropblock is performed, and SCT indicates the condition when complete LGSP dimensional dropblock is performed.

is more challenging than the Jester. On the other hand, from the results, we note that the regularization performance of the drop methods based on the guidance mechanism are generally better than the random-based drop methods, that verified our assumption.

C. Efficacy of Three Masks

To verify the contribution of each mask of three dimensions, we compare the performance of each individual mask and their combination in Table III. Specifically, the improvement effect is similar and effective when using the temporal and spatial masks alone, but the improvement effect is not very obvious when using the channel mask alone, which indicates that the spatial and temporal information in the action recognition task is the most important. When using the spatial and channel masks in combination, the effect improvement can reach a comparable level of TargetDrop[16], and when using all three in combination as we do in Algorithm 1, the effect reaches the best, top-1 accuracy increases 0.64% , and it can be observed that all three masks are indispensable in the action recognition task.

D. Analysis of Hyperparameters and Structure Selection

In this subsection, we further analyse the selection of hyperparameters (i.e., θ, γ, β, μ) in LGST-Drop and the way the combination of LGST-Drop with the network. As it is extremely time-consuming to train the whole Something-Something V2 dataset from scratch, we apply LGST-Drop to ResNet-50-TSM with extensive experiments on SS-V2-12 IV-A. As is shown in Fig.3-(a) and (b), our method is insensitive to different dropsize and keep probability, that may be due to the guided mechinism, this situation is consistent with [16], [15]. The keep probability of 0.85 and block size of 7 are slightly better. For the other two parameters, $1/\mu$ denotes the regular ratio on the channel dimension. $1/\beta$ denotes the regular ratio on the temporal dimension. It can be observed that the best performance is achieved with $\mu = 8$ and $\beta = 2$ in Fig.3-(c), and the effect decreases when it is greater or less than these values, which exactly indicates the necessity of regularization on channel and spatial dimension.

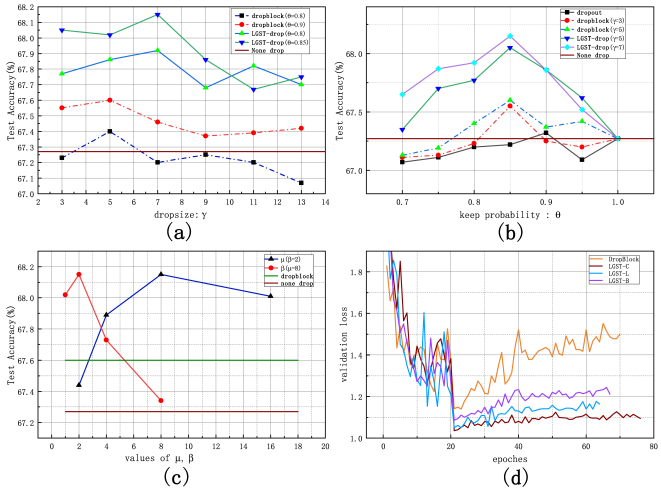

Fig. 3: Analysis of hyperparameters and structure Selection. In subfigure (a) and (b), we constrain $\mu = 8$, $beta = 2$. In subfigure (c), we constrain keeprate $\theta = 0.85$, droprate $\gamma = 3$. In subfigure (d), **DropBlock** denotes the validation loss curve of the ResNet-50-TSM which is trained on SS-V2-12 dataset by applying dropblock to convolution in layer-3 and layer-4. **LGST-C** denotes the curve of that applying LGST-Drop after every convolution in the network. **LGST-L** denotes the curve of that applying LGST-Drop after the four layers in the network. **LGST-B** denotes the validation loss curve of that applying LGST-Drop after every block (i.e., basicBlock, bottleneck).

On the other hand, in order to analyze the effect of the combination of our method with the network, we examined three different combinations: applying LGST-Drop to each convolution, each block (i.e. BasicBlock, Bottleneck), and all the four layers in ResNet50[1]. Figure 3-(d) shows the loss curves of the network on the validation set of SS-V2-12 for the combinations, and it can be seen that combining our method directly with the convolution get the best results, which indicates that the guideline-based drop strategy can alleviate the over-regularization situation.

E. Activation Visualization

In this subsection, we utilize Gradient-weighted Class Activation Mapping (Grad-CAM)[49] to visualize the activation units of ResNet-50-TSM on a video clip as shown in Fig.4. We can notice that the activation map generated by the model regularized with our method LGST-Drop demonstrates strong competence in capturing the extensive and relevant features towards the main object and frames. Compared with the other methods, the model regularized with LGST-Drop tends to precisely focus on more discriminative regions and more frames for action recognition which we attribute to aiming and masking certain valid features that correspond to critical channels and temporal frames.

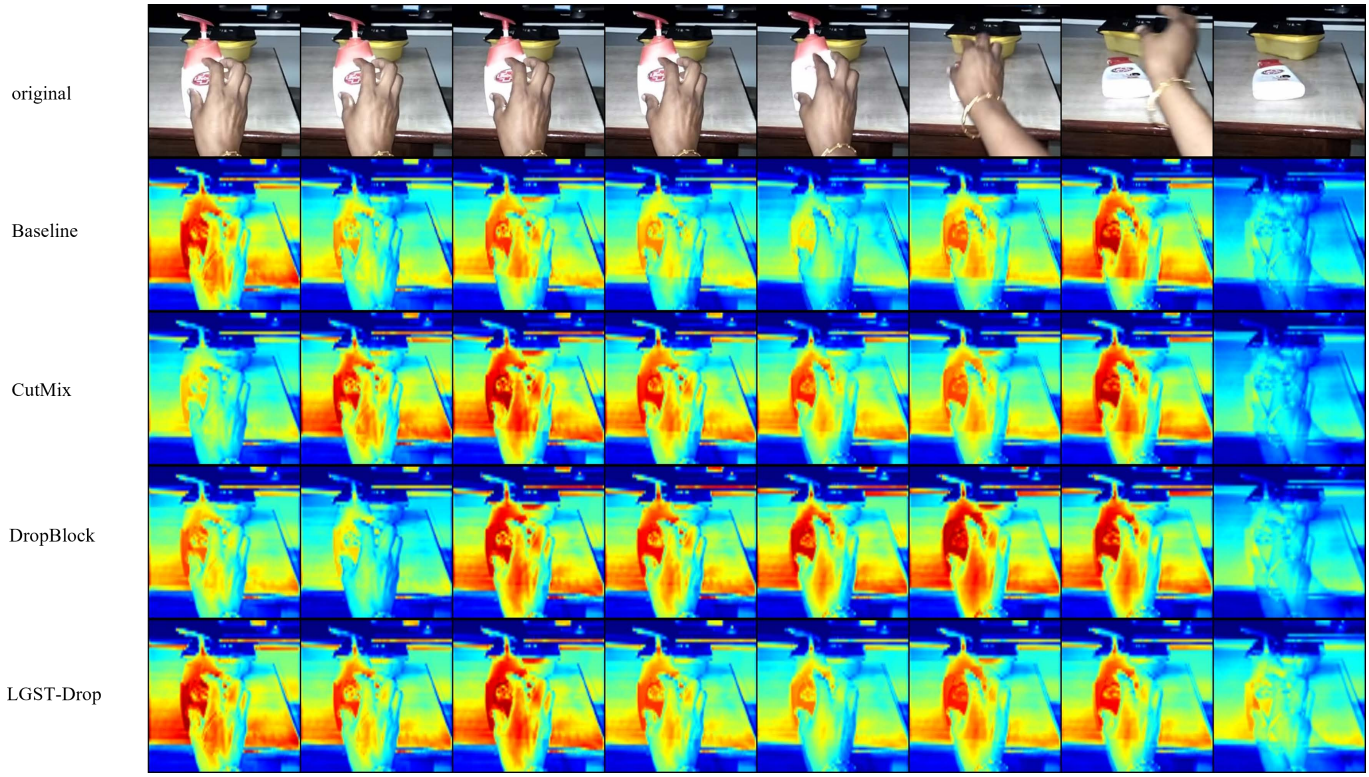

Fig. 4: Gradient-weighted Class Activation Mapping (Grad-CAM)[49] of "poking plastic bottle so that it falls over" for ResNet50-TSM trained without regularization (Baseline), with CutMix[22], DropBlock[14], and our LGST-Drop. The model trained with LGST-Drop has more activated units and can get more motion information from more frames.

V. Conclusion

In this paper, we propose an original regularization method LGST-Drop for spatiotemporal features and action recognition tasks. To deal with the problem of unstable regularization due to random dropping of previous dropout methods, LGST-Drop uses the Temporary Labels to guide dropping spatiotemporal regions, which improves the stability of the dropping efficiency. To deal with the problem of dimensional inequality in spatiotemporal models, we design spatial and temporal dimensional dropping strategies respectively. We have trained different datasets using LGST-Drop and compared them with other regularization methods, and the results show that the regularization of our method is significantly better for the action recognition task. We also visualize our method to verify it can better utilize the valid information of the input data in both temporal and spatial dimensions. However, it should be noted that the input data of this method must be video data, because LGST-Drop Mask is consisted of three Masks of different dimensions, so this method cannot be directly applied to other non-video tasks, but its ideology of dropping guided by the Temporary Labels mechanism is also worth exploring for other tasks.

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